

1. Simplify the expression: $F = A \cdot B + A \cdot B' \cdot C + A \cdot B' \cdot C'$.

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|---------------------|-----------|
| (A) A | (B) A + B |
| (C) $A \cdot B + C$ | (D) B + C |

2. What is the simplified SOP form for the function $F(A, B, C) = \sum m(0, 2, 4, 6)$?

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|----------|-------------------|
| (A) C' | (B) A' |
| (C) B' | (D) $A' \cdot C'$ |

3. Using the Consensus Theorem, simplify: $F = A \cdot B + A' \cdot C + B \cdot C$.

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|------------------------------|------------------------------|
| (A) $A \cdot B + A' \cdot C$ | (B) $A' \cdot C + B \cdot C$ |
| (C) $A \cdot B + B \cdot C$ | (D) $A + B + C$ |

4. How many essential prime implicants (EPIs) are in the K-Map of $F(A, B, C, D) = \sum m(0, 2, 5, 7, 8, 10, 13, 15)$?

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| (A) 2 | (B) 3 |
| (C) 4 | (D) 1 |

5. Express the Boolean function $F(X, Y) = X \oplus Y$ in canonical POS form.

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|------------------------|------------------------|
| (A) $(X + Y)(X' + Y')$ | (B) $(X + Y')(X' + Y)$ |
| (C) $X'Y + XY'$ | (D) $(X' + Y')(X + Y)$ |

6. Simplify: $F = (A + B) \cdot (A + B') \cdot (A' + C)$.

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|------------------|---------------------|
| (A) $A \cdot C$ | (B) $A + B \cdot C$ |
| (C) $A' \cdot B$ | (D) $A + C$ |

7. What is the complement of the expression $F = (A + B' \cdot C) \cdot D' + E$?

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|--|---------------------------------------|
| (A) $[(A' \cdot (B + C')) + D] \cdot E'$ | (B) $(A' + B \cdot C') \cdot D + E'$ |
| (C) $(A' + B \cdot C') + D \cdot E'$ | (D) $A' \cdot B \cdot C \cdot D + E'$ |

8. If $F(A, B, C) = \sum m(1, 3, 6, 7)$, what is its equivalent Maxterm representation $\prod M$?

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|---------------------------|---------------------------|
| (A) $\prod M(0, 2, 4, 5)$ | (B) $\prod M(1, 3, 6, 7)$ |
| (C) $\prod M(0, 1, 2, 3)$ | (D) $\prod M(4, 5, 6, 7)$ |

9. Which expression represents the dual of $(A + B') \cdot (C + 0) = A \cdot C + B' \cdot C$?

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| (A) $A \cdot B' + C \cdot 1 = (A + C) \cdot (B' + C)$ | (B) $(A' \cdot B) + (C' \cdot 1) = A' + B$ |
| (C) $(A + B') \cdot C = A \cdot C + B' \cdot C$ | (D) $A' \cdot B + C \cdot 0 = (A' + C) \cdot (B + C)$ |

10. Simplify the expression: $F = A'B'C' + A'BC' + AB'C' + ABC'$.

- (A) C' (B) C
(C) $A' + B'$ (D) $A + B$

11. A 4-variable K-Map grouping of 8 adjacent cells containing 1s reduces to a product term of how many variables?

- (A) 1 variable (B) 2 variables
(C) 3 variables (D) 4 variables

12. What is the output of the circuit implementing $F = (A \oplus B) \oplus (A \oplus B)$?

- (A) 0 (B) 1
(C) $A \oplus B$ (D) $A \cdot B$

13. Minimize $F(A, B, C) = \sum m(1, 3, 4, 5, 7)$ using a Karnaugh Map.

- (A) $C + A \cdot B'$ (B) $A + B \cdot C$
(C) $C' + A \cdot B$ (D) $B' + A \cdot C$

14. What is the Boolean expression for the output of a 2-to-1 Multiplexer with inputs $I_0 = A$, $I_1 = B$, and Select line $S = C$?

- (A) $A \cdot C' + B \cdot C$ (B) $A \cdot C + B \cdot C'$
(C) $(A + C) \cdot (B + C')$ (D) $A \cdot B + C$

15. Simplify the expression: $F = (A + B + C) \cdot (A + B + C')$.

- (A) $A + B$ (B) $A + C$
(C) $B + C$ (D) $A \cdot B$

16. Find the minimal SOP for $F(W, X, Y, Z) = \sum m(0, 1, 2, 8, 9, 10)$.

- (A) $X'Y' + X'Z'$ (B) $W'X' + Y'Z'$
(C) $X'Y' + W'Z'$ (D) $W'Y' + X'Z'$

17. Simplify $F = A \cdot B + A \cdot (B + C)' + B \cdot (B + C)'$.

- (A) $A \cdot B + A \cdot C'$ (B) $A + B$
(C) $A \cdot C'$ (D) $B \cdot C'$

18. Given $F = X'Y'Z + X'YZ + XY'Z + XYZ$, what is the reduced expression?

- (A) Z (B) $X' + Y$
(C) $X \cdot Y$ (D) Z'

19. How many NOR gates are required to realize an 2-input XOR function?

- (A) 4 (B) 3
(C) 5 (D) 2

20. Simplify the expression: $F = A'B + AB' + AB$.

- (A) $A + B$ (B) $A \cdot B$
(C) $A' + B'$ (D) $A \oplus B$

21. In a K-map with don't care conditions $d(A, B, C)$, don't care terms can be used as:

- (A) Either 0 or 1 to form larger groups (B) Strictly 1 only
(C) Strictly 0 only (D) Inverted inputs only

22. Convert $F(A, B) = A + B$ into standard Canonical Minterm form $\sum m$.

- (A) $\sum m(1, 2, 3)$ (B) $\sum m(0, 1, 2)$
(C) $\sum m(0, 2, 3)$ (D) $\sum m(1, 3)$

23. Simplify $F = (P + Q') \cdot (P' + Q)$.

- (A) $P \odot Q$ (XNOR) (B) $P \oplus Q$ (XOR)
(C) $P \cdot Q$ (D) $P' + Q'$

24. According to Shannon's Expansion Theorem, any function $F(X, Y, Z)$ can be expanded as:

- (A) $X' \cdot F(0, Y, Z) + X \cdot F(1, Y, Z)$ (B) $X + F(0, Y, Z) \cdot F(1, Y, Z)$
(C) $F(X, 0, 0) + F(0, Y, Z)$ (D) $X' \cdot [F(1, Y, Z)]'$

25. Simplify: $F = (A + B + C') \cdot (A + B + C) \cdot (A + B' + C)$.

- (A) $A + B \cdot C$ (B) $A \cdot B + C$
(C) $A' + B$ (D) $B + C$

ANSWER KEY

Q.No	Ans	Q.No	Ans	Q.No	Ans	Q.No	Ans	Q.No	Ans
1	(A)	6	(A)	11	(A)	16	(A)	21	(A)
2	(A)	7	(A)	12	(A)	17	(A)	22	(A)
3	(A)	8	(A)	13	(A)	18	(A)	23	(A)
4	(A)	9	(A)	14	(A)	19	(A)	24	(A)
5	(A)	10	(A)	15	(A)	20	(A)	25	(A)